RUNGTA CYBER SECURITY CLUB // END CHALLENGE

CTF-00 FIRST SHELL

TARGET your own Kali VM · **INTERFACE** the terminal only · **FLAGS** 5 core + 1 boss · **INSTRUCTOR** Aryan (Gorky)

THE BRIEFING. You just built a box from nothing: hypervisor, Kali, first boot. Now prove you can actually *live* in the shell. Six flags are hidden on your machine. No GUI, no file manager, no peeking at the setup script. Just you and the command line. Flag format: `RCSC{...}`. Write each one down as you capture it.

START HERE

Run the deploy command your instructor gives you, then read your orders:

```
$ bash ctf_setup.sh
$ cat ~/ctf-00/README
```

WARM-UP · NO FLAGS, JUST ORIENT YOURSELF

Answer these out loud before you hunt. Every one is a single command from today's session.

- Who are you logged in as? · What's your current directory?
- What kernel is this box running? · What's your IP address?
- How much disk is free? · How much RAM is in use right now?

tools: **whoami** · **pwd** · **uname -a** · **ip a** · **df -h** · **free -h**

CORE FLAGS · EVERYONE CLEARS THESE

FLAG 1 // FIRST CONTACT — CORE

Your orders point into `~/ctf-00`. Walk into the `intel/` folder, list what's there, and read the briefing file. The flag is sitting in plain text... if you can navigate to it.

needs: **cd** **ls** **cat**

FLAG 2 // NOTHING HIDDEN — CORE

There's a second asset in `intel/` you can't see with a plain `ls`. On Linux, files starting with a dot are hidden. Make them show themselves.

needs: **ls -la** (hidden / dotfiles)

FLAG 3 // LOCKED OUT — CORE

Try to read `~/ctf-00/loot/vault.txt`. Permission denied. Run `ls -l` on it and read the permission string on the left... it tells you exactly what's wrong. You own the file, so you're allowed to change its permissions. Grant yourself read access, then open it.

needs: **ls -l** (read perms) **chmod** (fix them) · see Field Manual

FLAG 4 // NEEDLE IN THE LOGS — CORE

The file `~/ctf-00/logs/auth.log` has 800 lines of noise and exactly one line with a flag. Do **not** scroll through it by hand. Search the file for the pattern `RCSC`.

needs: **grep**

FLAG 5 // GHOST IN THE PROCESS — CORE

Something is running on your box that shouldn't be. List every running process, filter for the tag `FIRST_SHELL_PROC`, and read the flag hiding in its command line. Then do the responsible thing and `kill` it by its PID.

needs: **`ps aux`** **`grep`** **`kill`**

BOSS FLAG · FOR THE ONES WHO WANT BLOOD

BOSS // BUILD THE VAULT — ADVANCED

Real admin work: you're going to provision access, not just read it. Build this exactly, using `sudo`:

- Create a group called `redteam`
- Create a user called `analyst`
- Add `analyst` to the `redteam` group
- Create the directory `~/ctf-00/restricted/ops`
- Set its owner to `analyst` and its group to `redteam`
- Set its permissions so owner and group have full access, everyone else has none (`770`)

When you think it's right, run `bash ctf_verify.sh`. If every check passes, it hands you the boss flag. If not, it tells you exactly what's still wrong.

needs: **`sudo`** **`groupadd`** **`useradd/adduser`** **`usermod -aG`** **`chown`** **`chmod`** · all in the Field Manual

FIELD MANUAL · YOUR ALLOWED TOOLKIT

COMMAND	WHAT IT DOES
`ls -la`	List all files incl. hidden, with permissions
`cd / pwd`	Move around / print where you are
`cat / less`	Read a file
`grep "x" f`	Find text `x` inside file `f`
`ps aux`	List every running process
`kill PID`	Stop the process with that PID
`man CMD`	The manual for any command

NEW TODAY	WHAT IT DOES
`sudo CMD`	Run CMD as root/admin
`chmod M f`	Change permissions (e.g. `chmod u+r f`, `chmod 770 f`)
`chown u:g f`	Change owner & group of `f` (needs sudo)
`groupadd g`	Create group `g`
`useradd -m u`	Create user `u` (or `adduser u`)
`usermod -aG g u`	Add user `u` to group `g`
`id u`	Show the groups user `u` is in

PERMISSION MATH (for FLAG 3 & BOSS). Each permission is a number: **read=4, write=2, execute=1**. Add them per group of three: owner / group / others. So `chmod 770` = owner `rwx` (7) + group `rwx` (7) + others `---` (0). `chmod 600` = owner `rw-`, nobody else. Read a file's current bits with `ls -l` — the 10 characters on the far left.

SCOREBOARD

☐ FLAG 1 ☐ FLAG 2 ☐ FLAG 3 ☐ FLAG 4 ☐ FLAG 5 ☐ **BOSS**

5/5 = you survived first boot. 6/6 = you own the box.